

README.md — Adventure Works Sales Dashboard

Project Overview

This project is an interactive business intelligence dashboard built in Microsoft Power BI using the **Adventure Works** dataset.

The dataset was originally provided in Microsoft Excel format and transformed into an analytical model using Power Query.

The goal of the project was to analyze:

- sales performance
- profitability
- customer behavior
- product efficiency
- reseller performance
- country-level trends

Data Preparation Process

Steps completed:

1. Data Import

Imported raw data from Excel into Power BI.

2. Data Transformation

Used Power Query for:

- cleaning null values
- formatting columns
- adjusting data types
- creating calculated columns
- preparing clean tables for modeling

3. Data Modeling

Built a relational model using **Star Schema** approach:

Fact Table 'Sales_data':

Contains transactional data:

- Sales Amount
- Product Cost
- Unit Price
- Customer IDs
- Product IDs
- Reseller IDs
- Dates

Dimension Tables:

- Customer_data
- Product_data
- Sales Territory_data
- Reseller_data
- SalesOrder_data
- Date_data

Relationships were created using key columns between fact and dimension tables.

4. DAX Measures Created

- Profit Margin %
- Average Order Value (AVO)
- Median Order Value
- Customer Count
- Reseller Count
- Purchase Frequency
- Growth vs Previous Year

5. Parameters Created:

- for decomposition tree: Profit, Sales Amount, Total Costs, Total Orders
- Top N Reseller
- Top N Customers



Dashboard Pages & Business Value

1 Overview Page

Main indicators:

- Sales Amount = **51.9M**
- Profit = **5.8M**
- Profit Margin = **11%**
- Orders = **23.5K**
- Customers = **17.6K**

Departments:

- Executive Management
- Finance
- Sales Department

Business Questions Answered:

- Is the company growing year over year?
- Are profits increasing with revenue?
- How many customers place orders?
- What sales channels perform better?

Recommendations:

- ✓ Continue scaling high-performing years like FY2020
- ✓ Improve margin, not only revenue
- ✓ Expand stronger sales channels (Internet = 92%)
- ✓ Monitor reseller performance separately

2 Product Analysis Page

Visualizations:

- Top products by orders
- Top products by profit
- Orders and Profit Margin by category
- Subcategory sales
- Unit Price vs Standard Cost
- Drill through the products (price and cost)

Departments:

- Product Team
- Procurement
- Marketing
- Sales

Business Questions:

- Which products sell the most?
- Which products generate the most profit?
- Which categories underperform?
- Are pricing levels optimal?

Key Insights:

- Bikes generate highest sales
- Accessories have high volume
- Top profitable products: Mountain series bikes

Recommendations:

- ✓ Increase stock of top-selling bikes
- ✓ Bundle accessories with bike purchases
- ✓ Review low-performing SKUs
- ✓ Optimize pricing based on cost margins

3 Country Analysis Page

Visualizations:

- Profit by country
- Margin by country
- Orders by country
- Drill-down by product/category/channel

Departments:

- Regional Management
- Finance
- Strategy Team

Business Questions:

- Which countries generate highest profit?
- Where are margins weak?
- Which channels work best in each country?

Key Insights:

Top markets:

1. United States
2. Australia
3. United Kingdom

Recommendations:

- ✓ Prioritize US and Australia growth investment
- ✓ Rework low-margin countries

4 Customer Analysis Page

Visualizations:

- Customers by country
- Orders vs AVO and Median OV by country
- Top customers by order count
- Customers by city map

Departments:

- CRM
- Marketing
- Sales

Business Questions:

- Where are most customers located?
- Which countries have highest order values?
- Who are repeat customers?
- Which cities have strong demand?

Recommendations:

- ✓ Launch loyalty campaigns for top buyers
- ✓ Focus acquisition in strong cities
- ✓ Run location-based marketing campaigns tailored to specific countries and cities to improve conversion rates and optimize ad spend
- ✓ Upsell in countries with high AVO

5 Reseller Analysis Page

Visualizations:

- Profit margin by city
- Orders vs margin by business type
- Orders vs margin by country
- Reseller city map

Departments:

- B2B Sales
- Partnerships
- Distribution Team

Business Questions:

- Which reseller types are most profitable?
- Which cities have strong reseller networks?
- Which partners reduce margin?

Recommendations:

- ✓ Focus on profitable reseller types
- ✓ Renegotiate low-margin partnerships
- ✓ Expand in profitable reseller cities
- ✓ Segment reseller programs by type



Strategic Business Recommendations

Revenue Growth

- Focus expansion on Bikes category
- Scale e-commerce / Internet sales channel
- Cross-sell accessories

Profitability

- Improve pricing strategy
- Reduce low-margin reseller deals
- Compare the top cities by customers and the top cities by sellers, and optimize the sellers

Customer Growth

- Retention campaigns
- VIP segmentation



Tools Used

- Microsoft Excel
- Microsoft Power BI
- Power Query
- DAX
- Data Modeling



Final Result

Developed a full interactive BI solution that helps stakeholders monitor the business indicators, identify profitable products, optimize markets, and improve strategic decision-making.

Portfolio Value

This project demonstrates skills in:

- Data Cleaning
- Power Query ETL
- Star Schema Modeling
- DAX Calculations
- Dashboard Design
- Business Analysis
- Storytelling with Data